SYSTEMATIC REVIEW

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Effectiveness of joint mobilization techniques on the range of motion of adult patients with primary adhesive shoulder capsulitis: systematic review and meta-analysis

The effectiveness of joint mobilization techniques for range of motion in adult patients with primary adhesive capsulitis of the shoulder: a systematic review and meta-analysis

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Summary

AIM

To determine the effectiveness of range-of-motion joint mobilization techniques in adult patients with primary adhesive shoulder capsulitis.

METHOD

Systematic review with meta-analysis. The search was carried out in the MEDLINE/PubMed, PEDro, CENTRAL, LILACS, EMBASE, CINAHL, Scopus and Web of Science databases. The eligibility criteria were studies that used an oscillatory and/or sustained joint mobilization technique applied, alone or added, to a treatment program in patients with primary adhesive capsulitis, at any stage. Study selection and data extraction were performed by two authors independently. The risk of bias was assessed according to the tool proposed by Cochrane.

RESULTS

14 studies with varying risks of bias were included. Posterior mobilization versus another joint technique does not present a significant average difference (- 0.95 degrees; 95% confidence interval; - 5.93 to 4.02), while compared to a control group the difference is 26. 80 degrees (95% confidence interval; 22.71 to 30.89), also when a set of joint techniques is applied versus a control group, for abduction the difference is 20.14 degrees (95% confidence interval; 10.22 to 30.05). In both cases, the results are statistically significant and a moderate effect size is seen.

CONCLUSIONS

The scientific evidence is not conclusive about the effectiveness of joint mobilization, given the heterogeneity in the methodology and bias of the articles included in this review. When compared

with treatments that do not include manual therapy, they seem to favor the improvement of range of motion and reduction of pain in patients with primary adhesive capsulitis of the shoulder.

Key ideas

There is controversy in the evidence about the effectiveness of joint mobilization techniques in patients with primary adhesive shoulder capsulitis.
Our results show that the evidence is inconclusive.
The included studies show high heterogeneity and risk of bias.

Introduction

Adhesive capsulitis is a common musculoskeletal condition, characterized by spontaneous onset pain associated with a progressive loss of glenohumeral movement of unknown etiology [1] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Neviaser was the first to use the term adhesive capsulitis, describing it as a chronic inflammation of the synovial membrane with fibrosis of the joint capsule and intra-articular adhesions in the shoulder [2] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Years later, it would be defined by the Consensus of the American Society of Shoulder and Elbow Surgeons as "a condition of unknown etiology, characterized by a significant restriction of the passive and active range of motion of the shoulder that occurs in the absence of intrinsic pathology known" [3] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Zuckerman proposed a classification scheme where primary or idiopathic adhesive capsulitis is not associated with a systemic condition or history of damage, that is, there is no preceding event to which the condition can be attributed. From a diagnostic point of view, it is considered in all cases that the etiology associated or underlying this condition cannot be identified [3] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [4] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

Knowledge of the natural history of the disease is relevant to evaluate the real effectiveness of the different therapeutic modalities used in the clinical management of this pathology. In the case of primary adhesive capsulitis, its benign nature and self-limiting course is controversial [5] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [6] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [7] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [8] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [9] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Codman and Grey were the first to state that all patients with primary adhesive capsulitis have complete recovery from pain and range of motion two years after symptoms began [10] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [11] (/webactivo/editor.cgi?id=6183&num=1&web=1#). However, prospective and long-term studies have reported varying percentages of range of motion loss, pain, and disability in patients with primary adhesive capsulitis [6] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [12] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [13] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [14] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [15] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Regarding movement restrictions, the authors report significant restriction in 90% of patients at seven months of follow-up [7] (/webactivo/editor.cgi?id=6183&num=1&web=1#), and between 30 to 50% of patients present mild and/or moderate restriction at a follow-up of three to 10 years [6], (/webactivo/editor.cgi?id=6183&num=1&web=1#) [7] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [8], (/webactivo/editor.cgi?id=6183&num=1&web=1#) [12] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [13] (/webactivo/editor.cgi?id=6183&num=1&web=1#). This background is relevant, even more so when moderate quality evidence suggests that the theory of self-resolution is increasingly uncertain and that the best

therapeutic results are obtained by intervening in early and not late stages [16] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

In this sense, it should be noted that rotational movements, specifically external rotation, have not shown significant changes before four weeks of physiotherapy treatment [17] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Some therapeutic interventions such as corticosteroid injection, different electrotherapy modalities, arthrographic distension, muscle trigger point treatment and manual therapy techniques have also been recommended to improve range of motion and reduce pain [18] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [19] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [20] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [21] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [22] (/webactivo/editor.cgi?id=6183&num=1&web=1#). In recent years, several studies have shown beneficial results using manual therapy techniques to correct glenohumeral rotational range of motion deficits, especially external rotation [23] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [24] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [25] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [26] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [27] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [28] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Such techniques have been shown to be effective in the management of capsular restraints, specifically when provided by a physical therapist [29] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

The literature describes three types of joint mobilization techniques: oscillatory, sustained and manipulative. Oscillatory techniques are repetitive passive movements of varying amplitude and at low speed [30] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [31] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [32] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Sustained mobilizations are a type of traction of variable amplitude and low speed, which reduces intra-articular compressive forces and eliminates distension of the periarticular tissues [33] (/webactivo/editor.cgi?id=6183&num=1&web=1#). These have been graduated according to the range of motion and the degree of tension of the periarticular tissue [31] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [34] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Finally, manipulation techniques are those that involve high speed and low amplitude movements associated with a *thrust* at the end of the available joint range of motion [31] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

From the above, our research questions arise: in patients with primary adhesive capsulitis of the shoulder, are sustained joint mobilization techniques more effective than oscillatory ones for improving range of motion, pain and function of the shoulder? Adding techniques Does joint mobilization to a treatment program produce any improvement in range of motion, pain, and shoulder function?

Trying to elucidate these questions was what motivated the carrying out of this systematic review, whose main objective is to determine the effectiveness of joint mobilization techniques in improving range of motion, whether applied alone or added to a treatment program. Secondary objectives are to determine the effect of the techniques on improving function and reducing pain, in addition to the dosage used.

Methods

Protocol

This systematic review will be reported according to the preferred regulations for systematic reviews and meta-analysis [35] (/webactivo/editor.cgi?id=6183&num=1&web=1#). It is worth mentioning that the protocol prior to carrying out this review was not recorded.

Eligibility criteria

To carry out the systematic review, a search strategy was developed in which the following eligibility criteria were included:

1. Population: patients over 18 years of age diagnosed with primary adhesive capsulitis at any stage, without distinction of gender or race.
2. Intervention: an oscillatory and/or sustained joint mobilization technique applied alone or added to a treatment program.
3. Comparison: with other manual therapy techniques, use of physical agents, different modalities of therapeutic exercise or pharmacological treatments.
4. Outcome measures: articles that have evaluated clinical effectiveness across range of motion, function, and shoulder pain.
5. Types of studies: randomized controlled clinical trials published in English or Spanish until July 16, 2018.

Inclusion criteria

Randomized controlled clinical trials comparing the clinical effectiveness between two joint mobilization techniques applied in patients with primary adhesive capsulitis.
Randomized controlled clinical trials comparing the clinical effectiveness of one or more joint mobilization techniques applied as part of a treatment program in patients with primary adhesive capsulitis.

Exclusion criteria

Randomized controlled clinical trials that do not describe the type of joint mobilization technique or specify the dose used.
Randomized controlled clinical trials studying other joint mobilization techniques such as mobilization with movement and/or manipulation.

Sources of information

MEDLINE/PubMed (www.ncbi.nlm.nih.gov/pubmed), PEDro (www.pedro.org.au), CENTRAL (www.cochrane.org), LILACS (www.lilacs.bvsalud.org/es), EMBASE (www.elsevier.com/solutions/embase-biomedical-research echa), CINAHL (www.health.ebsco.com/products/the-cinahl-database echa), Scopus (www.scopus.com/home.uri?zone=header&origin=searchbasic) and *Web of Science*(www.fecyt.es/es/resource/web-science), searching from the beginning of each database until July 16, 2018. Reference lists of included trials and any relevant review articles retrieved in the electronic search were also reviewed, with the aim of identifying any other potentially relevant trials. It is worth mentioning that a search for gray literature was not performed in the present review.

Search strategies

To carry out the search in the MEDLINE/PubMed database, a sensitive search strategy proposed in the Cochrane Collaboration manual was used [36] (/webactivo/editor.cgi?id=6183&num=1&web=1#). The search terms used in this review were obtained from MeSH: *bursitis, physical therapy modalities and musculoskeletal manipulations*. These terms were combined with the free text terms: frozen shoulder, adhesive capsulitis and manual therapy.

The process is described below:

1. Bursitis[MeSH]
2. *Frozen shoulder*
3. *Adhesive capsulitis*
4. #1 OR #2 OR #3
5. *Physical therapy modalities*[MeSH]
6. *Musculoskeletal manipulations*[MeSH]
7. *Iherapy manual*
8. #5 OR #6 OR #7
9. #4 AND #8
10. *Randomized controlled trial*[pt]

11. *Controlled clinical trial*[pt]
12. *Randomized*[tiab]
13. *Randomly*[tiab]
14. Trial[tb]
15. *Groups*[tiab]
16. #10 OR #11 OR #12 OR #13 OR #14 OR #15
17. *Animals[*mh] *NOT humans*[mh]
18. #16 NOT #17
19. #9 AND #18

For PEDro, LILACS, CENTRAL, CINAHL databases *Web of Science*, Scopus and EMBASE the search strategy was carried out by combining the terms previously mentioned in the advanced search option.

Selection of studies

Two of the authors (JZ and FP) performed the search string independently, reviewing the title, abstract, and full text of the articles. Only studies that met the eligibility criteria were selected. The disagreements were resolved in a discussion between the authors. If disagreement persists, a third independent reviewer (CO) was consulted.

Data extraction

Data extraction for the included trials was carried out as recommended in the Cochrane collaboration manual for systematic reviews [36] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Two reviewers (JZ and FP) independently completed a standardized data summary form, specially designed for this review. The data collected was entered into RevMan 5.3 and, in case of disagreement or discrepancy, the article was evaluated by a third reviewer (HG). Its final inclusion was decided through discussion by the authors.

Risk of bias assessment of included studies

Two reviewers (JZ and FP) independently assessed the risk of bias of the included studies, using the Cochrane collaboration tool for risk of bias, as described in the manual for systematic reviews [36] (/webactivo/editor.cgi?id=6183&num=1&web=1#). The following domains were considered:

1. Random sequence generation.
2. Concealment of assignment (selection bias).
3. Blinding of participants and staff (achievement bias).
4. Blinding of the evaluation of results (detection bias).
5. Incomplete outcome data (desertion bias).
6. Selective reporting of results (report bias).
7. Other sources of bias, each of these domains was assessed as “low risk”, “high risk”, or “unclear risk”.

A study classified as high risk would not be excluded from this review, but could degrade our confidence of recommendation. The information obtained was used through a multiple analysis presentation, based on all the studies obtained and that allowed their quantitative analysis.

Summary measures

For the present review, only continuous variables were identified (range of movement, pain scales and function), so the difference in means between the intervention groups with 95% confidence intervals was used as the main summary measure, in order to determine the effect size of the results. To carry out this process, the final value and standard deviation of the outcome of interest were recorded, as well as the number of participants for each treatment group at the end of follow-up.

Synthesis of results

The synthesis of the results was carried out through meta-analyses, which were carried out in the RevMan 5.3 program. In the case of clinically homogeneous studies, heterogeneity was evaluated with the Chi-square statistical test and the heterogeneity I^2 test, we consider a Chi-square value with a $P > 0.1$ and the I^2 test to be low heterogeneity with a value less than or equal to 50%. In this case, a fixed effect model was used for data analysis, with inverse variance weighting. In contrast, a random effect model was used when there was high heterogeneity.

Additional analysis

We performed the meta-analysis based on two comparisons. First, we pooled randomized controlled clinical trials that included subsequent grade III mobilization according to Kaltenborn. Secondly, the studies that included a set of joint mobilization techniques within their intervention were analyzed. In both comparisons, subgroup analysis was considered based on comparison with other joint mobilization techniques and with conventional treatment that did not include joint mobilization techniques.

Results

Selection of studies

According to the defined search strategy, 563 records were identified for the terms and keywords, of which 225 were duplicates. Once the titles and summaries of the remaining articles were analyzed, only 25 articles met the eligibility criteria, which were reviewed in full text. After analyzing the selection criteria, 11 studies were excluded by different criteria, all of them presented in Figure 1, and the remaining 14 studies were included in the qualitative synthesis process

[37] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[38] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[39] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[40] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[41] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[42] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[43] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[44] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[45] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[46] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[47] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[48] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

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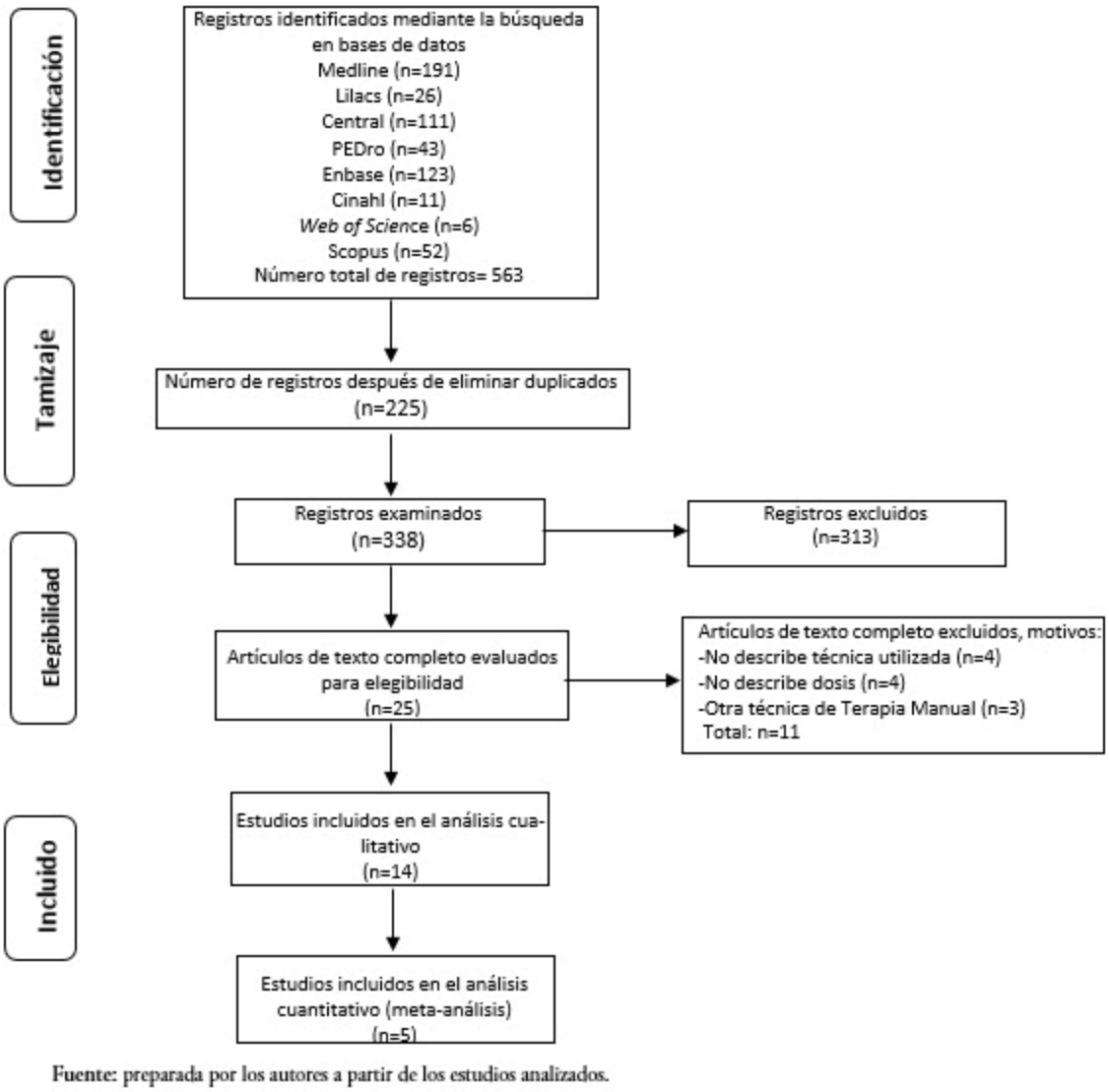
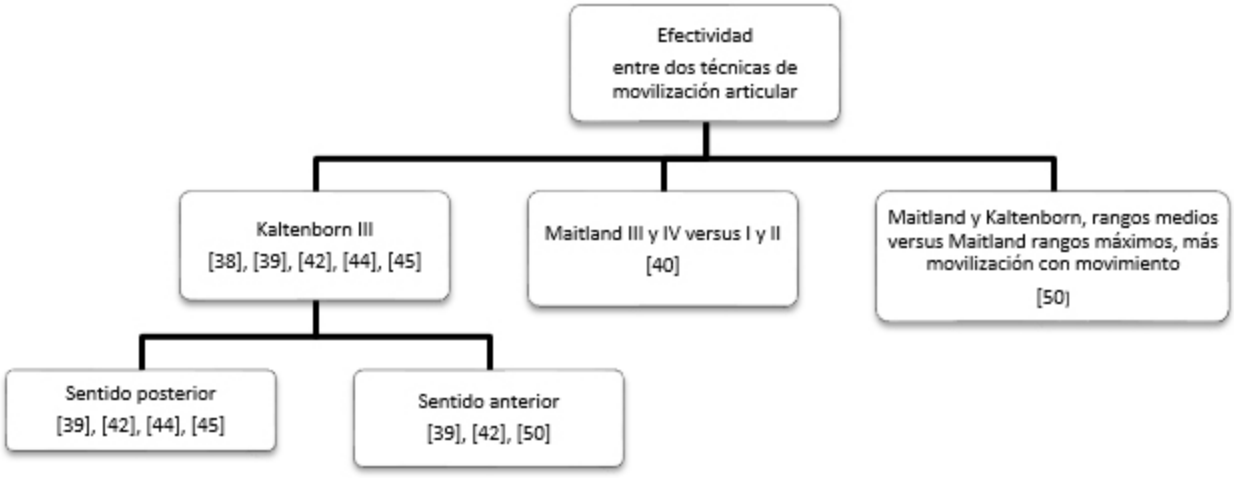


Figure 1. Flowchart of the phases of the systematic review

Characteristics of the studies

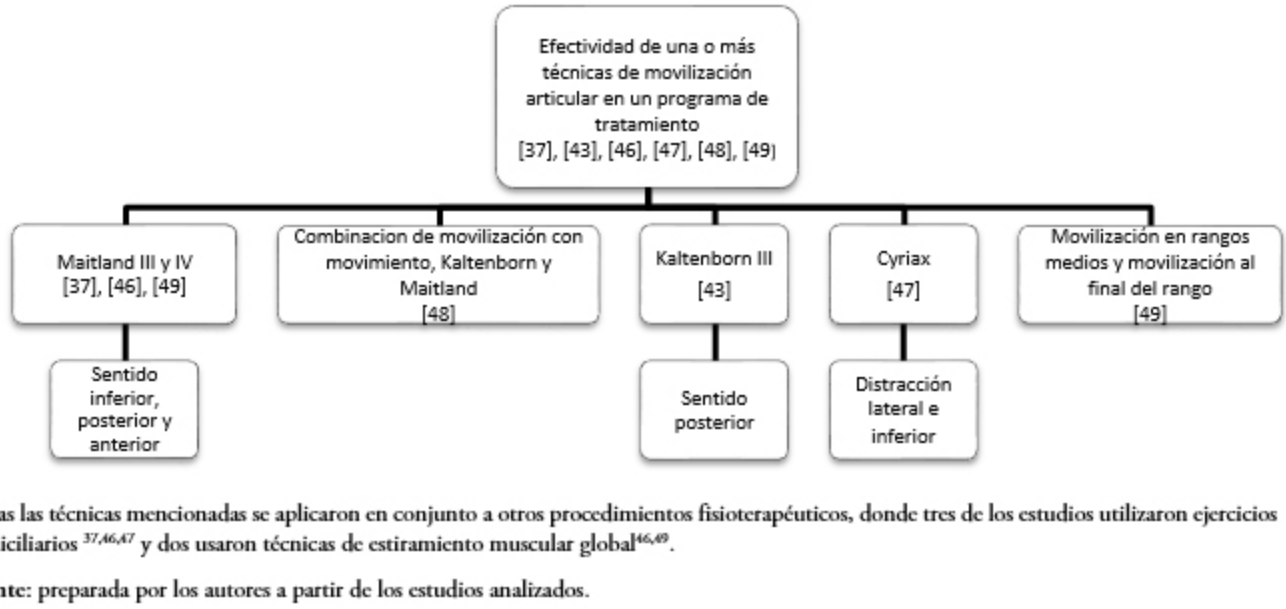
Considering the 14 selected randomized controlled clinical trials, the total number of patients was 474, with an average of 24 patients per study and a range between 14 and 100 patients. The average age in the studies was 46.9 years, with a range between 35 and 70 years.

After analyzing the included articles, the joint mobilization techniques studied could be grouped based on two comparisons: those that compared the clinical effectiveness between two joint mobilization techniques [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [40] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [41] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [42], (/webactivo/editor.cgi?id=6183&num=1&web=1#) [44] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [50] (/webactivo/editor.cgi?id=6183&num=1&web=1#), and those that compared the clinical effectiveness of one or more joint mobilization techniques applied as part of a treatment program [37] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [47] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [48] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [49] (/webactivo/editor.cgi?id=6183&num=1&web=1#). The schematic representation of these comparisons can be seen in Figures 2 and 3.



Fuente: preparada por los autores a partir de los estudios analizados

Figure 2. Studies that compared the effectiveness between two joint mobilization techniques.



Todas las técnicas mencionadas se aplicaron en conjunto a otros procedimientos fisioterapéuticos, donde tres de los estudios utilizaron ejercicios domiciliarios [37,46,47] y dos usaron técnicas de estiramiento muscular global[46,49].

Fuente: preparada por los autores a partir de los estudios analizados.

Figure 3. Comparative studies on one or more joint mobilization techniques applied in a treatment program.

Results of individual studies

The results of each of the included studies are summarized in Table 1.

Risk of bias of studies

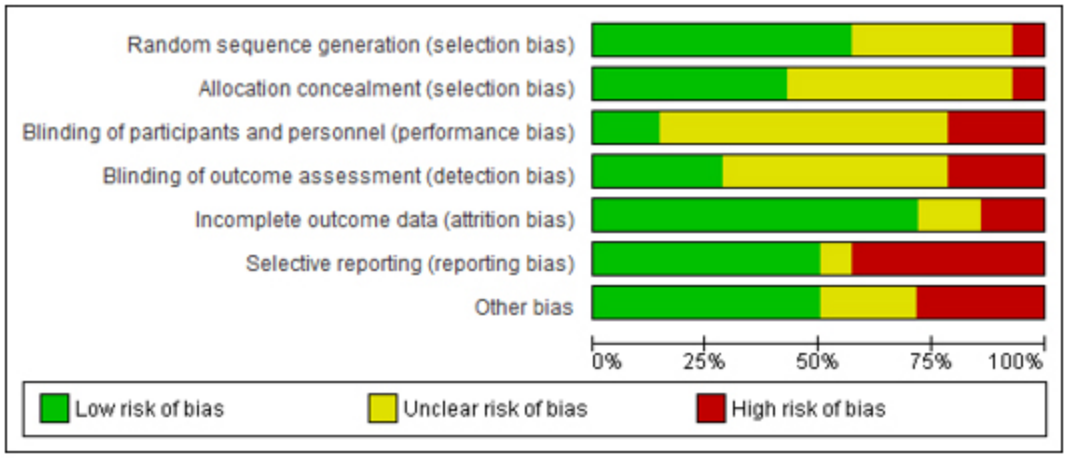
The risk of bias of the studies included in this review is presented in Figures 4 and 5. Only eight of the selected studies reported details of randomization [40] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [42] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [44] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [47] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [50] (/webactivo/editor.cgi?id=6183&num=1&web=1#), and the assignments were hidden in only six [41] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [44] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [47] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Given the nature of the interventions studied, blinding of patients and treaters is complex. At this point three studies were classified as high risk [37] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#) and the rest of the risk is unclear.

Autor/ Año	Población	Intervención	Comparación	Medidas de resultado
Sharad et al[37] 2011	N = 22 ♀10♂12 40-60 años MDA: No especificada GC.: n = 11 47,4 años (DE 5,49) GE.: n = 11 46,5 años (DE 4,44)	5 x semanas casa GE: US + terapia física GH + M III y IV 10 a 15 repeticiones + Ejercicios en casa	GC: US + terapia física + Ejercicios en casa	Tras 3 semanas: -RDM ABD, RI, RE, FLEX: ↑GE >↑GC p ≤ 0,05 -Dolor (EVA): ↓GE =↓GC p = 0,44
Agarwal et al[38] 2016	N =22 ♀13♂15 40-70 años MDA: No especificada GC: n=15 48,7 (DE 6,4) GE: n =15 52,5 (DE 96)	3X semanas GE: Técnica de distracción inversa 10-15 repeticiones + terapia convencional + ejercicios en casa para RDM	GC: KIII lateral x 1 min + 15 min estiramiento + deslizamiento posterior + deslizamiento caudal + terapia convencional +ejercicios en casa para RDM	Tras 6 semanas: -RDM: • ABDa y ABDp: ABDa y ABDp ↑GE > ↑GC p = 0,01 • REa y REp: REa y REp GE = GC p > 0,05 -Funcionalidad: • HBB: GE = GC p = 0,29 • FLEX-SF: ↑GE > ↑GC
Sirajuddin et al[39] 201	N = 45 ♀28♂17 40-60 años MDA: no especificada. GC: n =15 GE1: n =15 GE2: n =15	GE1: KIII anterior sostenido x 1min /15 min + terapia física GE2: KIII posterior sostenido x 1min /15 min + terapia física	GC: Terapia física	Tras 12 sesiones: -Dolor (EVA): (↓GE1>↓GE2) > ↓GC p = 0,02 -RDMa • RE: ↑GE1 > ↑GE2 p = 0,00; ↑GE2 > ↑GC p = 0,01; ↑GE1 > ↑GC p = 0,00 • RI: ↑GE1 > ↑GE2 p = 0,02; ↑GE2 > ↑GC p = 0,01; ↑GE1 = ↑GC p = 1 • ABD: ↑GE1 = ↑GE2 p = 1; ↑GE2 > ↑GC p = 0,00; ↑GE1 > ↑GC p = 0,01 1. -Funcionalidad (SRQ): (GE1 Y GE2)>GC
Vermeulen et al[40] 2006	N = 100 ♀66♂34 MDA: GNA GC: n = 49 51,6 años (DE 7,6) GE: n = 51 51,7 años (DE 8,6)	2x semanas / 12 semanas GE: M III y IV según tolerancia + FNP + Codman	GC: M I y II según tolerancia + FNP + Codman	Tras 12 meses: -RDM: • ABDa y ABDp: ↑GE > ↑GC p = 0,59 y p = 0,52 • REa y REp: ↑ GE > ↑ GC p = 0,51 y p < 0,01 • Flexa y Flexp: ↑ GE > ↑ GC p = 0,38 y p = 0,54 -Dolor (EVA) • Reposo: ↓ GE > ↓ GC p = 0,66 • Movimiento: ↓ GE > ↓ GC p = 0,34 • Nocturno: ↓ GE > ↓ GC p = 0,18 -Funcionalidad: -SRQ y SDQ: ↑ GE > ↑ GC p = 0,49 y p = 0,03 -SF-36: • Físico: ↑ GE > ↑ GC p = 0,79 • Mental: ↑ GE > ↑ GC p = 0,34
Sarkari et al[41] 2006	N = 20 ♀9♂11 40-65 años MDA: No especificada GC: n = 10 56,1 años (DE 4,95) GE: n = 10 58,3 años (DE 4,37)	3x semanas GE: M III – IV anteroposterior 10 a 15 repeticiones, 5 a 6 veces x 20 min + CHC + MP + elongación posterior y anterior de capsula.	GC: M III – IV inferior 10 a 15 repeticiones, 5 a 6 veces x 20 min + CHC+ MP + elongación posterior y anterior de capsula.	Tras 9 semanas -RDMp ABD: ↑GC > ↑ GE p = 0,00 -RDMa ABD: ↑ GC > ↑ GE p = 0,00 -Dolor y discapacidad (SPADI): ↓GC = ↓GE p = 0,16
Harsimram et al[42] 2011	N = 15 ♀6♂9 35-70 años MDA: aleatorización en bloque. GC: n = 8 52 años (DE 14,1) GE: n = 7 56 años (DE 4,3)	5 sesiones GC: KIII anterior 5 repeticiones de 30 seg + 1 min movimiento pasivo + CHC + Codman + ejercicios en muralla. GE: KIII posterior 5 repeticiones de 30 seg + 1 min MP + CHC + Codman + ejercicios en muralla	GC: KIII anterior 5 repeticiones de 30 seg + 1 min movimiento pasivo + CHC + Codman + ejercicios en muralla	No especificado tiempo -RDM ABD, RI, RE -Dolor (EVA) Ambas técnicas resultan efectivas en el aumento de la rotación externa de hombro
Gutiérrez et al[43] 2015	N=57 ♀46♂11 50-58 años MDA: GNA GC: n = 28 53,3 años (DE 4,4) GE: n = 29 58,8 años (DE 4,7)	2-3 x semanas /10 sesiones GE: KIII posterior 15 repeticiones X 1 min + cicloergómetro EESS.	GC: US + Codman + ejercicios con bastón + isometría.	Tras 1 mes -RDM: • RE: ↑GE > ↑GC p = 0,00 • Flex anterior.: ↑GE > ↑GC p = 0,00 • ABD: ↑GE > ↑GC p = 0,00 -Dolor (EVA): ↓GE > ↓GC p = 0,00 -Funcionalidad (Score de Constant-Murley): ↑GE > ↑GC p = 0,00
Johnson et al[44] 2007	N = 18 ♀16♂4 37-66 años MDA: G.N.A GC: n = 10 54,7 años (DE 8) GE: n = 8 50,4 años (DE 6,9)	2-3 x semanas GE: KIII posterior sostenida por 1 min con un total de 15 min TTO + US + distracción axial + cicloergómetro.	GC: KIII anterior sostenida por 1 min con un total de 15 min TTO + US + distracción axial + cicloergometro.	A las 3 semanas -Dolor (EVA): ↓GC = ↓GE p = 0,31 -RDM: • REa: ↑ GC < ↑ GE p = 0,00 -Funcionalidad (cuestionario de función auto asistida): ↓ GC = ↓ GE p = 0,36
Do Moon et al[45] 2015	N = 20 MDA: sobres sellados GC: n = 10 48,3 años (DE 2,98) GE: n = 10 49,1 años (DE 3,07)	3 semanas/ 12 sesiones GE: KIII posterior 15 repeticiones de 30 seg cada una con pausas de 10 seg + CHC + TIF	GC: M III 1 oscilación por seg total de 15 por 10 min + CHC + TIF	Tras 1 mes -Dolor (EVA): ↓GE = ↓GC p > 0,05 -RDMp • RE: ↑ GE = ↑ GC p > 0,05 • RI: ↑ GE = ↑ GC p > 0,05
Celik et al[46] 2015	N = 26 ♀18♂8 MDA: Aleatorización computarizada GC: n = 14 58,8años (DE 6,4) GE: n = 12 54,2 años (DE 7,9)	18 sesiones x 6 semanas GE: M (I y II las 2 primeras semanas y III y IV las últimas 4 semanas) inferior, posterior y anterior + stretching + Programa de ejercicio domiciliario.	GC: stretching + Programa de ejercicio domiciliario.	Tras 1 año -Dolor (EVA): ↓GE = ↓GC p = 0,60 -Funcionalidad: • DASH: ↓GE = ↓GC p = 0,55, tras 1 año p = 0,03 • Score de Constant -Murley: ↑GE > ↑GC p = 0.04 -RDMp • Flex: ↑GE = ↑GC p = 0.14 • ABD: ↑GE > ↑GC p = 0,00 • RE: ↑GE > ↑GC p = 0,02 • RI: ↑GE = ↑GC p = 0,09
Chan et al[47] 2010	N = 14 ♀11♂3 MDA: Aleatorización computarizada. GC: n =7 56,7 años (DE 6,6) GE: n = 7 50,9 años (DE 10,3)	GE: Ejercicios domiciliarios + Cyriax (distracción lateral e inferior) + Cyriax movilización pasiva de flexión con distracción	GC: Infiltración previa, ejercicios domiciliarios.	Tras 10 semanas -Dolor (EVA): ↓GC = ↓GE -Funcionalidad • SPADI: ↑GC = ↑GE -RDMp • RE: ↑GC = ↑GE • RI: ↑GC = ↑GE • ABD: ↑GC = ↑GE
Park et al[48] 2014	N = 53 ♀40♂13 56 años (DE 7,6) MDA: no se especifica	2 x semanas GE1: Kenalog + lidocaína + M + K + Mulligan GE2: M + K + Mulligan GE3: Kenalog + lidocaína + distensión capsular Todos recibieron 20 minutos de terapia que incluía CHC + TENS + US	GC: 20 minutos de terapia que incluía CHC + TENS + US	A las 4 semanas -Funcionalidad • SPADI y Constant Murley: ↑GE1 > ↑GE2 = ↑GE3 > ↑GC p < 0,05 -RDMa FLX, ABD, RE RI (cinta métrica): ↑GE1 > ↑GE2 = ↑GE3 > ↑GC p < 0,05, excepto GE3 en RE (p > 0,05) -Dolor (EVA): ↓GE1 > ↓GE2 = ↓GE3 > ↓GC p < 0,05
Yang et al[49] 2012	N = 34 ♀22♂11 MDA: azar para GC y sobres sellados al resto de los grupos GC: n = 11 54,3 años (DE 7,6), sujetos con menor restricción del movimiento.	2 x semanas /8 semanas GE: Movilización al final del rango M IV (10-15 repeticiones, con 30s descanso) + movilización escapular.	GC, GC1: Movilización en rango medio, técnicas de stretching + fisioterapia (US, OC y/o electroterapia) y ejercicios activos	Tras 3 meses -RDM p (inclinómetro) • RI: ↑GE > ↑GC1↑GC p = 0,0005 • RE: ↑GE > ↑GC1↑GC p = 0,002 -Discapacidad (FLEX-SF): ↑GE > ↑GC1 > ↑GC p = 0,005 -Cinemática (FASTRAK) • Tipping escapular: ↑GE > ↑GC1↑GC p = 0,004

	GC1: n = 12 54,9 años (DE 10,3) GE: n = 12 56,8 años (DE 7,2)			• Ritmo ET: ↑GE > ↑GC1 ↑GC p = 0,002
Yang et al[50] 2007	N = 28 ♀24♂4 MDA: G.N.A GC: n = 14 53,3 años (DE 6,5) GE: n = 14 58 años (DE 10,1)	GE: A-C-A-B Dónde A: MRE 10 a 15 repeticiones hacia la restricción. B: MRM en 3 direcciones al final del movimiento 10 – 15 repeticiones C: MWM 3 series de 10 repeticiones	GC: A-B-A-C	Tras 12 semanas -Discapacidad (FLEX – SF) • -Cinemática del hombro: FASTRAK. • Elevación hombro • Ritmo escapular -RDM (RI y RE) B y C p < 0,01, A p > 0,05 en FLEX-SF, FASTRAK y RDM Entre grupos B y C p > 0,05 excepto en ritmo escapular, donde B obtuvo un mayor cambio con p>0.05

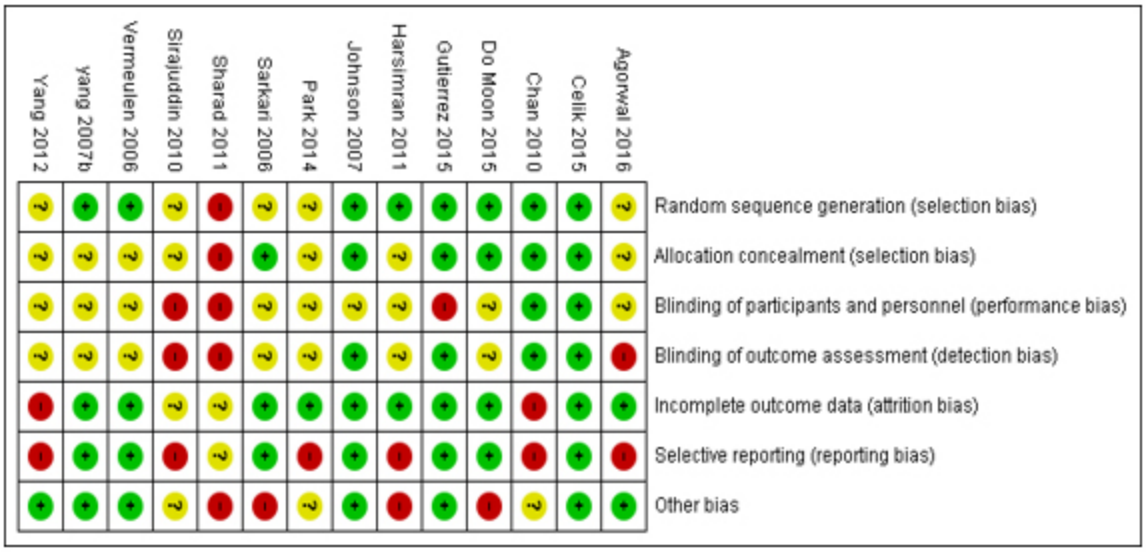
Abreviaturas: ABDa o p= Abducción activa o pasiva, CHC= Compresas Húmedo Caliente, DS= Desviación Estándar, DASH= Disabilities of the Arm, Shoulder and Hand, EVA= Escala Visual Análoga, EESS= Extremidad Superior, ET= Escapulo Torácico, FLEXa o p= Flexión activa o pasiva, FLEX-SF= Flexilevel Score of Shoulder Dysfunction, FASTRAK= Sistema de Análisis de Movimiento, FNP= Facili-tación Neuromuscular Propioceptiva, GC= Grupo Control, GC1= Grupo Control1, GE= Grupo Experimental, GE1, 2...= Grupo Experimental 1, 2..., GH= Glenohumeral, GNA= Grupo No Aleatorizado, HBB= Hand Behind Back, K= técnicas Kaltenborn, M= técnicas Maitland, min= minutos , MDA=Medida De Aleatorización, MP= Movilización Pasiva, MRE= Movilización en Rango Extremos, MRM= Movilización en Rango Medio, MWM= Movilización con Movimiento, N= muestra total, n= muestra, OC= Onda Corta, p= p-valor, RDMa o p= Rango de Movimiento activo o pasivo, RIa o p= Rota-ción Interna activa o pasiva, REa o p= Rotación Externa activa o pasiva, SF-36= 36-item Short-From Health Survery, SRQ= Shoulder Rating Questionnaire, SDQ= Shoulder Disability Questionnaire, SPADI= Shoulder Pain and Disabiliti Index, seg= segundos, TTO= Tartamiento, TIF= Terapia Inerferemcial, US= Ultrasonido.

Table 1. Features of the included items.



Fuente: obtenido a través de *Review Manager* (RevMan) [*Computer program*]. Version 5.3. Copenhagen: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014*

Figure 4. Risk of bias graph presented as a percentage of all included articles.



Fuente: obtenido a través de *Review Manager* (RevMan) [*Computer program*]. Versión 5.3. Copenhague: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014.*

Figure 5. Summary of the risk of bias assessment of the included articles.

Measurements of study results

Considering that motion restriction is a pathognomonic sign of primary adhesive capsulitis, for the present systematic review range of motion was considered the primary outcome measure. All included studies evaluated glenohumeral range of motion through goniometry. Special emphasis was placed on osteokinematic movements of external rotation and shoulder abduction. Considering the maladaptive and pathophysiological changes of primary adhesive capsulitis on tasks involving the shoulder joint complex

[38] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[40] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[41] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[43] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[44] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[46] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[50] (/webactivo/editor.cgi?id=6183&num=1&web=1#), shoulder function and pain were considered secondary outcome measures. The first was evaluated in 10 of the studies, through different questionnaires such as Constant-Murley [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), *The Disabilities of the Arm, Shoulder and Hand* [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), *Hand behind back* [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), *Shoulder Pain and Disability Index* [41] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[46] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[47] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[48] (/webactivo/editor.cgi?id=6183&num=1&web=1#), among others. Pain was evaluated through the visual analogue scale in 10 of the 14 studies

[37] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[39] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[40], (/webactivo/editor.cgi?id=6183&num=1&web=1#)

[42] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[43] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[44] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[45] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[46] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[47] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[48] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

Meta-analysis of the results

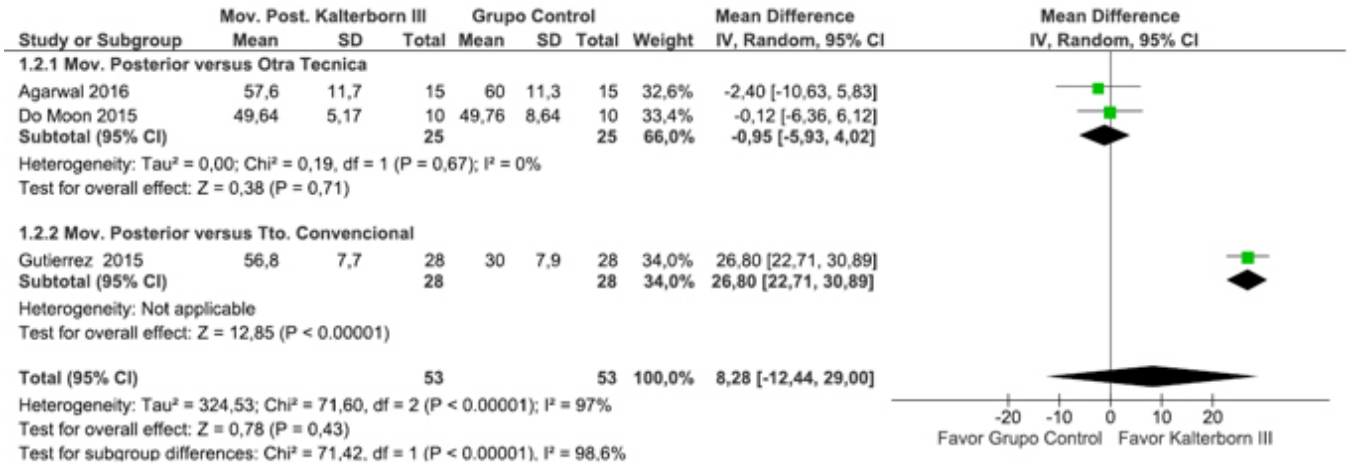
Glenohumeral range of motion

Three of the included articles evaluated glenohumeral external rotation

[38] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

[43] (/webactivo/editor.cgi?id=6183&num=1&web=1#),

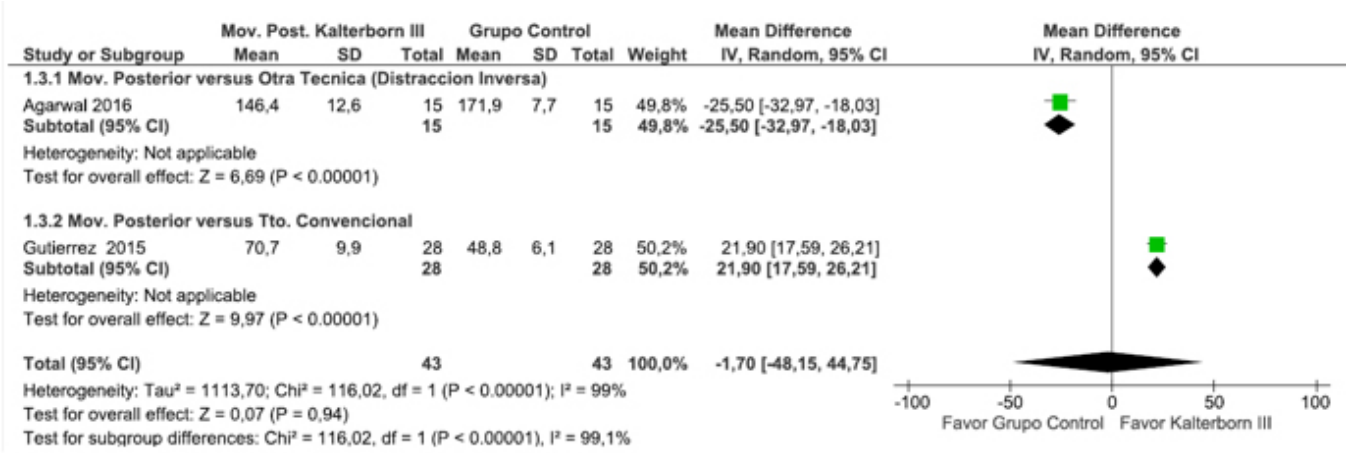
[45] (/webactivo/editor.cgi?id=6183&num=1&web=1#). When comparing posterior mobilization versus other joint techniques, Figure 6 shows that the results of the meta-analysis with a fixed effect model, there was no statistically significant difference in the range of motion of external rotation at the end of the treatment mean difference of -0.95 (95% confidence interval: -5.93 to 4.02; p = 0.71). When comparing posterior mobilization versus conventional treatment, there was a statistically significant difference in favor of posterior mobilization, showing an average difference of 26.8 degrees (95% confidence interval: 22.71 to 30.89, p = < 0.05). When considering the total effect with high heterogeneity (I^2 = 97%) no significant difference (p = 0.43) could be established in external rotation in favor of subsequent mobilization.



Fuente. Obtenido a través de *Review Manager* (RevMan) [*Computer program*]. Versión 5.3. Copenhague: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014.*

Figure 6. Comparison between posterior Kaltenborn mobilization versus other treatments for glenohumeral external rotation range.

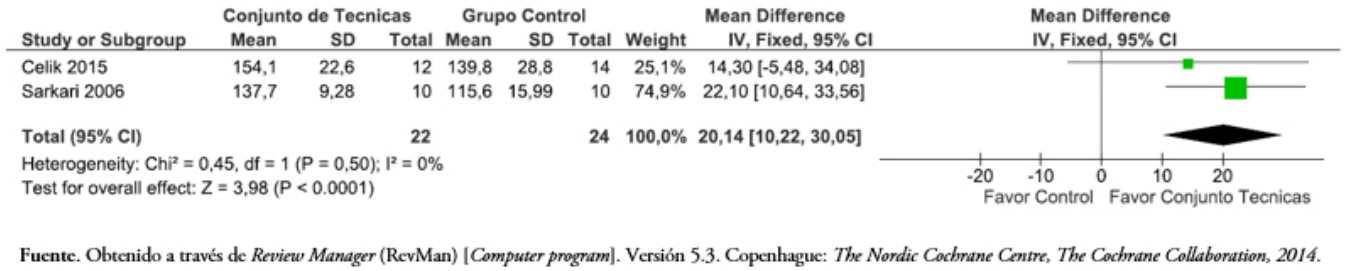
Two of the included articles evaluated glenohumeral abduction [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Comparing posterior mobilization versus inverse distraction, Figure 7 shows that there was a statistically significant difference in favor of posterior mobilization with a mean difference of 25.5 degrees (95% confidence interval: 32.97 to 18.03; $p < 0.05$). And when compared with a conventional treatment, the mean difference is 21.9° (95% confidence interval: 17.59 to 26.21) in favor of subsequent mobilization ($p < 0.05$). When considering the total effect with a random effect model due to high heterogeneity (I^2 = 99%), no significant difference ($p = 0.94$) could be established in abduction in favor of subsequent mobilization.



Fuente. Obtenido a través de *Review Manager* (RevMan) [*Computer program*]. Versión 5.3. Copenhague: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014.*

Figure 7. Comparison between subsequent Kaltenborn mobilization versus other treatments for glenohumeral abduction range.

When comparing a treatment program that included joint mobilization techniques versus a control group, Figure 8 shows that with a fixed effect model there was a statistically significant difference of 20.14 degrees (95% confidence interval: 10.64 to 33.56) in the range of glenohumeral abduction in favor of studies that included a set of joint techniques within their intervention compared to a control intervention ($p < 0.05$).

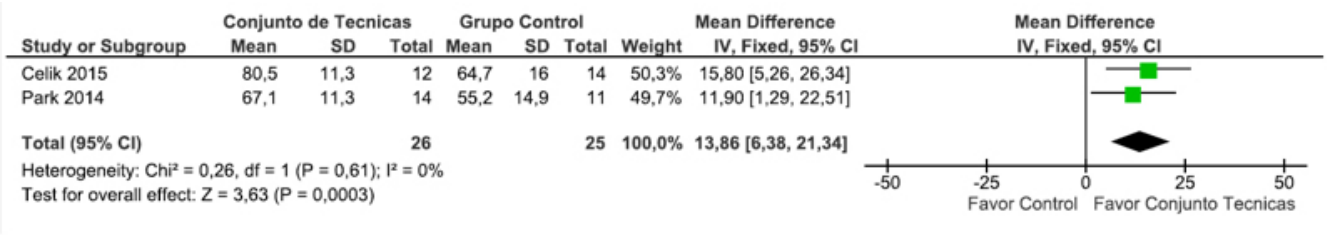


Fuente. Obtenido a través de *Review Manager* (RevMan) [*Computer program*]. Versión 5.3. Copenhague: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014.*

Figure 8. Comparison between a treatment program with joint mobilization techniques versus control group for glenohu-meral abduction range.

Shoulder function

Two of the included articles evaluated shoulder function with Constant-Murley, [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [48] (/webactivo/editor.cgi?id=6183&num=1&web=1#). When comparing a treatment program that included joint mobilization techniques versus a control group, Figure 9 shows that with a fixed effect model there was a statistically significant difference of 13.86 points (95% confidence interval: 6.38 to 21.34) in the Constant-Murley questionnaire in favor of the joint use of joint techniques compared to a control intervention ($p < 0.05$).

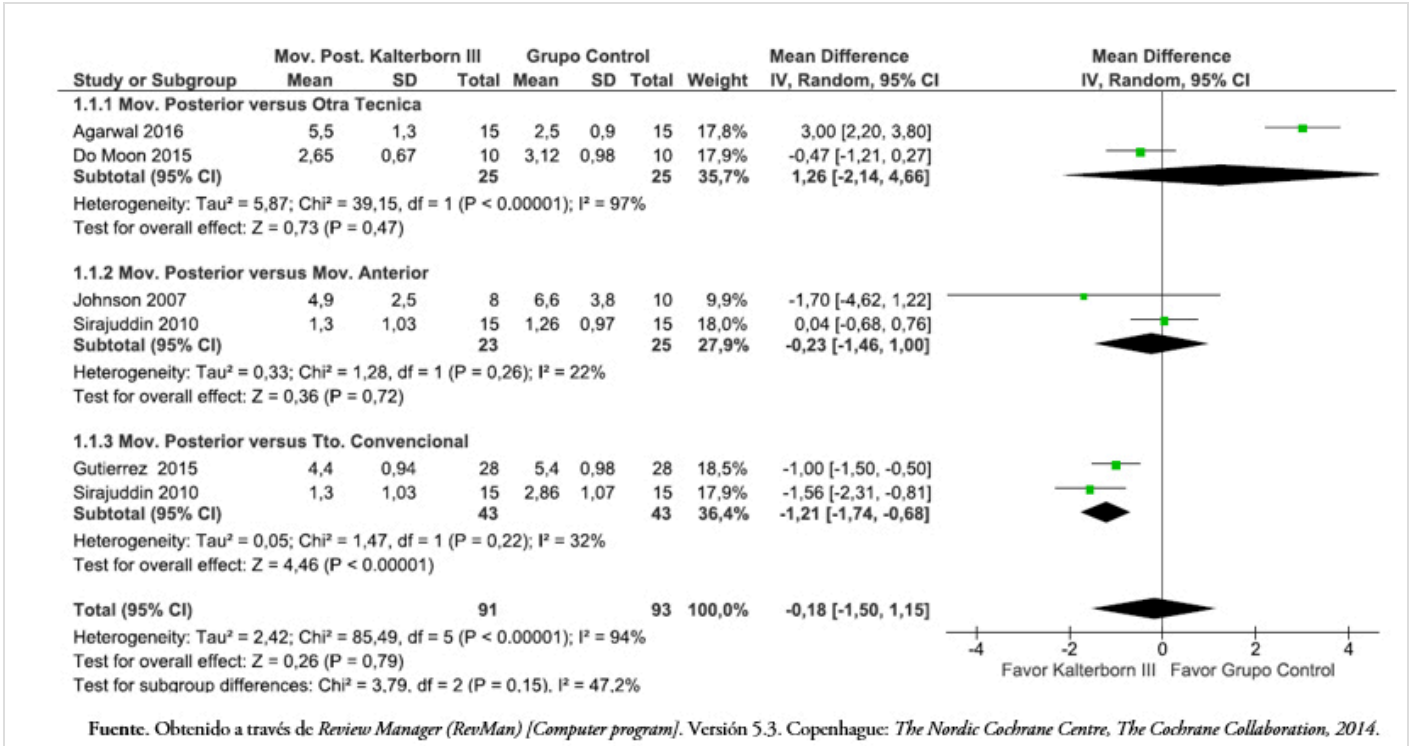


Fuente. Obtenido a través de *Review Manager (RevMan) [Computer program]*. Versión 5.3. Copenhague: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014*

Figure 9. Comparison between a treatment program with joint mobilization techniques versus a control group for the Constant-Murley questionnaire.

Pain

Five of the included articles assessed pain intensity with the visual analogue scale [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [44] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#). When comparing posterior mobilization versus other joint techniques, Figure 10 shows that, with a random effect model due to high heterogeneity, there is an average difference of 1.26 centimeters (95% confidence interval: -2.14 to 4.66), and versus mobilization in the anterior direction it is -0.23 centimeters (95% confidence interval: -1.46 to 1). In both comparisons, the difference was not statistically significant (p > 0.05). In contrast, when compared to treatments that do not include joint mobilization techniques, there was an average difference of -1.21 centimeters (95% confidence interval: -1.74 to -0.68) in favor of mobilization. later (p < 0.05).



Fuente. Obtenido a través de *Review Manager (RevMan) [Computer program]*. Versión 5.3. Copenhague: *The Nordic Cochrane Centre, The Cochrane Collaboration, 2014.*

Figure 10. Comparison between subsequent Kaltenborn mobilization versus other treatments for pain intensity with a visual analogue scale.

Discussion

The present systematic review with meta-analysis was conducted based on 14 randomized controlled clinical trials; and aimed to determine the effectiveness of joint mobilization techniques applied alone or added to a treatment program in improving range of motion, function and shoulder pain. The results show that the evidence is not conclusive in the use of joint mobilization techniques, despite the high heterogeneity and risk of bias of the included articles, the Kaltenborn type III technique shows a tendency to increase the range of external rotation, abduction and pain reduction compared to a control group, although without statistically significant differences.The use of joint mobilization techniques added to a treatment program show statistically and clinically significant results for shoulder abduction and function when compared with control groups, in patients with primary adhesive capsulitis.

According to previous systematic reviews that have evaluated the effectiveness of joint mobilization techniques, the results have shown an increase in range of motion and a decrease in pain [22] (/webactivo/editor.cgi?id=6183&num=1&web=1#). When compared with conventional treatments, they have shown moderate effect changes in shoulder range of motion, especially in external rotation ($p < 0.05$), in favor of posterior sliding, showing an average increase of 26.8 degrees (95% confidence interval: 22.71 to 30.89). Although this difference does not occur when comparing two joint mobilization techniques ($p = 0.71$). Considering the high heterogeneity of the studies, it cannot be assured that these changes depend only on the technique, given that it has not been possible to identify a single, minimum or necessary dosage to obtain these improvements. Howeverthe most relevant clinical results were obtained in those experimental groups that were subjected to high-degree mobilization techniques and performed at the end of the available range of motion [37] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [49] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [50] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

Considering that the restriction of rotational movement is one of the most common characteristics of the condition, this deficit can last beyond the restrictive phase and persist for a period close to two years, affecting external rotation more than other movements of the shoulder [51] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Restrictions of periarticular tissue [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [44] (/webactivo/editor.cgi?id=6183&num=1&web=1#) together with the consequent dysfunction of the muscles that stabilize and mobilize the glenohumeral and scapulothoracic joints, they have their origin in increased tension in the rotator interval and directly affect the joint complex of the shoulder in all its planes of movement [52] (/webactivo/editor.cgi?id=6183&num=1&web=1#). Changes in the translational and rotational movement of the humeral head determine a pathomechanical movement characterized by an ascent and anteriorization of the humeral head with respect to the glenoid surface, affecting its arthrum and osteokinematics. The above could explain why humeral rotations end up being the most restricted movements in conjunction with abduction, which does not always show statistically significant changes [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [40] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [47] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

Despite the existing knowledge and the large number of published trials, it has not been possible to identify or establish a dosage for the techniques found, either as a single treatment or as part of an intervention program. When analyzing complementary treatments to the applied techniques, a study that added stretching techniques to grade IV Maitland joint mobilization [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), showed favorable results in both groups, with significant changes in favor of joint mobilization in the range of motion, a similar situation observed in another study that added Kaltenborn III and Maitland III techniques to the application of physical agents [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), which showed important changes for the two groups but without statistically significant differences between them. In those studies that used Maitland techniques (in all grades) associated with an exercise program [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), and Maitland III and IV techniques associated with muscle stretching [41] (/webactivo/editor.cgi?id=6183&num=1&web=1#), statistically significant improvement was shown in abduction range of motion. Case similar to what was presented in a study that added scapular mobilization, and that in addition to improving the abduction range of motion, also showed improvement in glenohumeral external rotation [49] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

The dosage of the techniques used in the different studies is varied, observing that the frequency of repetitions of the same technique to complete a treatment session, in the case of Kaltenborn techniques, ranges from five to 15 repetitions, with a maintenance time from 30 seconds to one minute [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#) showing positive changes in the intervened groups. The total application time of the technique was 15 minutes, except for one study that considered 20 minutes [48] (/webactivo/editor.cgi?id=6183&num=1&web=1#). On the other hand, those who used Maitland's techniques describe 2 to 15 repetitions, the frequency of which was mentioned in only two studies [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [46] (/webactivo/editor.cgi?id=6183&num=1&web=1#), which range from one to two oscillations per second, with a total treatment time of 20 minutes. Two of the included studies propose 10 to 15 repetitions for joint mobilization techniques, but do not indicate which specific technique this dosage corresponds to [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [50] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

Regarding the direction of the maneuver, the authors who used Kaltenborn techniques studied it anteriorly, posteriorly and laterally [40] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [42] (/webactivo/editor.cgi?id=6183&num=1&web=1#) demonstrating that in those studies where the technique was used posteriorly, the improvement in the range of motion of external rotation was statistically significant [38] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [44] (/webactivo/editor.cgi?id=6183&num=1&web=1#). This contrasts with the results of the study by Do Moon and collaborators [45] (/webactivo/editor.cgi?id=6183&num=1&web=1#), where no changes of statistical importance were reported when comparing Maitland grade III techniques in different senses.

High-grade joint mobilization techniques, whether Maitland III - IV or Kaltenborn grade III, aim to restore arthrokinematic movement through distension and elongation of periarticular structures, increasing the tension of some capsular and ligamentous components, that passively stabilize the joint [53] (/webactivo/editor.cgi?id=6183&num=1&web=1#). This concept refers to the physical components of the length-tension curve, which studies the behavior of a tissue when it is subjected to a load, demonstrating that its properties vary, progressing from an elastic phase to a plastic phase, depending directly on the degree, time and dosage of mobilization, as well as the relative position of the joint [54] (/webactivo/editor.cgi?id=6183&num=1&web=1#). The remodeling of collagen fibers and the modulation of the properties of the affected connective tissue depend on this [50] (/webactivo/editor.cgi?id=6183&num=1&web=1#). When analyzing studies that used techniques in the final range and high degree of posterior tension, all obtained improvement in range of motion, especially external rotation, but not for abduction (p = 0.94).

This postulates that the magnitude of the technique and the direction in which it is performed can determine a greater or lesser change, where posterior sliding influences capsular restriction to a greater degree, positioning the joint and increasing the elongation of the joint capsule, which which would not be enough to show a determining effect on the abduction movement product of the elevation of the humeral head and the increase in the thickness of the lower fibers of the joint capsule [39] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [40] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [47] (/webactivo/editor.cgi?id=6183&num=1&web=1#). However, joint mobilization techniques achieved significant improvements in the abduction range when compared to control groups, achieving an average increase of 20.14 degrees (95% confidence interval: 10.64 to 33.56).

One of the characteristics to consider, and that only one study recorded in its inclusion criteria [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), was the degree of irritability of the subjects. Kelley et al [1] (/webactivo/editor.cgi?id=6183&num=1&web=1#), they propose a classification system in this regard, where a low degree of irritability is one that presents pain less than or equal to three centimeters according to the analogous visual scale, without pain at night or at rest, and with similar limitation of the active range of motion and passive. On the other hand, a high degree of irritability implies suffering from pain greater than or equal to seven centimeters of analogous visual scale, mainly during passive mobility, with pain at night and at rest, associated with high levels of disability, with pain predominating over restriction of the range of motion.

This finding becomes relevant when assigning a treatment, since some patients will not tolerate the degree of mobilization if their level of irritability is high, mainly because tolerance to mechanical loads is low. Since high-degree mobilization techniques are based on micro damage [37] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#), it is likely that when they are performed, the clinical manifestation is being negatively influenced, reflected in an increase in pain. Its phase-dependent applicability could be a topic of study, supporting its application in phases where stiffness predominates over pain [40] (/webactivo/editor.cgi?id=6183&num=1&web=1#), [43] (/webactivo/editor.cgi?id=6183&num=1&web=1#).

Limitations

The trials identified for this review studied a wide variety of joint mobilization techniques, this added to the risk of bias and heterogeneity they presented limited the possibility of grouping a greater number of results. Furthermore, the great variability that the studies presented in terms of the dosage of their interventions (type of technique, number and frequency of sessions) made it difficult to identify optimal doses for the application of any joint mobilization in patients with primary adhesive shoulder capsulitis. Within methodological limitations, despite using eight electronic databases and reference tracking of relevant studies, some studies may have been overlooked.Searching for gray literature was not considered and only studies in English or Spanish were selected. We were also unable to assess the risk of publication bias due to the limited number of trials.

Conclusion

In summary, 14 randomized controlled clinical trials were analyzed that used joint mobilization techniques, with variable methodology in relation to potential sources of bias and statistically heterogeneous. When applying these techniques alone or as part of a treatment program, compared to treatments that do not include joint mobilization, they appear to improve range of motion, function, and decrease pain.

There was no evidence to prove or disprove a clinical difference of any type of technique over another, nor is there sufficient evidence to determine the ideal dose of the different joint mobilization techniques used in the different randomized controlled clinical trials in patients with primary adhesive capsulitis.

Notes

Declaration of authorship

JZG: Conceptualization, Methodology, Data Validation, Formal Analysis, Research, Materials, Data Management, Writing (Preparation of the original draft), Writing (Review and editing), Review and presentation, Supervision, Project Management.
FPB: Conceptualization, Writing (Preparation of the original draft), Writing (Review and editing), Review and presentation.
HGE: Conceptualization, Methodology, Formal analysis, Writing (Preparation of the original draft), Writing (Review and editing), Review and presentation.
COH: Software, Data validation, Formal analysis, Data management, Writing (Preparation of the original draft), Writing (Review and editing).

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The authors have completed the ICMJE conflict of interest declaration form, and declare that they have not received financing to prepare the report; not have financial relationships with organizations that could have interests in the published article, in the last three years; and not have other relationships or activities that could influence the published article. The forms can be requested by contacting the responsible author or the editorial management of the *Magazine*.

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